

Lesson Exemplar in General Science

Quarter 3

Lesson

10

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Bureau of Learning Delivery
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LESSON EXEMPLAR

Learning Area	General Science	Grade Level	11
Semester	1	Quarter	3

I. OBJECTIVES *(Identifying the Goals)*

Content Standard	The learners learn that climate change affects ecosystems and the organisms living in them; and ecosystems are finite and have limited resources.
Performance Standard	By the end of the quarter, learners use examples to describe the relationships among living systems and their components and how the structure of a part relates to its function. They explain the role of cells in perpetuating life and how organ systems work together to maintain optimal health. They conduct investigations on the effects of climate change on local ecosystems.
Learning Competencies	The learners explain how climate change impacts Philippine ecosystems;
II. REFERENCES and MATERIALS (Selecting Resources and Material)	“Born to Be Wild: Effects of Deforestation in the Philippines.” 2019. Www.youtube.com. https://youtu.be/em1eRDsXD4U. Diving Deep into Philippine Environmental Protection: The DENR’s Crusade for a Greener Archipelago! - TaasNooPilipino. (2024, December 8). TaasNooPilipino - Kahit Kanino. https://taasnoopilipino.com/diving-deep-into-philippine-environmental-protection-the-denrs-crusade-for-a-greener-archipelago/ Meih. 2025. “Climate Change and the Philippines.” Slideshare. 2025. https://www.slideshare.net/slideshow/climate-change-and-the-philippines-1315236/1315236.



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	“Philippines: Changing Climate Changing Lives.” n.d. Www.youtube.com. https://youtu.be/VvmPRrJC5cc. PAGASA. 2023. “Climate Change in the Philippines.” PAGASA. 2023. https://www.pagasa.dost.gov.ph/information/climate-change-in-the-philippines. Roberto Vagabond. 2022. “Philippines’ Most Successful Mangrove Reforestation Project!” YouTube. September 3, 2022. https://www.youtube.com/watch?v=53zh0ZnY0g. Villaroza, S. M. (2025). Philippine ecosystems lecture notes. Slideshare. https://www.slideshare.net/slideshow/philippine-ecosystems-lecture-notes/43899693 2025. Youtu.be. 2025. https://youtu.be/y19WEesgy2k.	
III. CONTENT <i>(Sequencing Content)</i>	Impact of Climate Change on Philippine Ecosystems	
IV. OBJECTIVES <i>(Setting Clear Objectives and Analyzing the Tasks)</i>	At the end of the lesson, the learners should be able to: 1. Identify the major ecosystems in the Philippines. 2. Explain the different ways climate change affects these ecosystems. 3. Describe and analyze real-world examples of climate change impacts in the country. 4. Explain the importance of interconnections between climate change, ecosystems, and human life.	
IV. PROCEDURES		ANNOTATION
A. Activating Prior Knowledge	The teacher initiates preliminary activities every day, such as prayer, attendance checking, unfreezing, and other classroom management activities.	



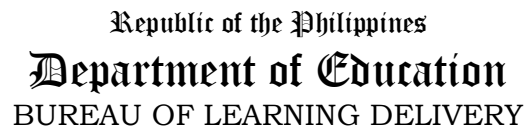
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	1. Activating Prior Knowledge OPTION 1: ECOSYSTEM PHOTO SORTING Students classify photos of forests, reefs, fields, etc. into categories. Procedure: 1. Divide the class into small groups (4–5 learners each). 2. Provide each group with a set of shuffled ecosystem photos. Instruct them to sort the images into correct categories (forest, coastal/marine, freshwater, agricultural, urban). 3. Groups post their sorted sets on Manila paper and briefly explain their choices. Assessment/Rubric (5 pts each = 20 total): Accuracy of Classification	<i>The teacher uses this activity to gauge learners' prior knowledge of Philippine ecosystems. It encourages group collaboration and discussion while allowing the teacher to identify misconceptions, such as confusing land use with natural ecosystems, which can be clarified before proceeding to the main lesson.</i> <i>Sample Key Question for Option 1 (after all the presentations): In summary, what do you know so far about the different ecosystems in the Philippines, and why is it important to distinguish them before discussing climate change impacts?</i> <i>Sample Key Question for Option 2 (after the activity): Based on the statements shared, what do you know so far about how climate change is affecting ecosystems in the Philippines?</i>
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	Group Participation Clarity of Explanation Organization of Output OPTION 2: COMMUNITY SHARING Learners describe changes they've seen in local environments (e.g., flooding, coral bleaching). Procedure:	<i>The teacher uses this activity to diagnose learners' prior knowledge and interests about ecosystems and climate change. It helps reveal misconceptions that can be clarified during the lesson while also aligning the discussion with learners' curiosity. The KWL chart further serves as a reflective tool when the L column is revisited at the end of the lesson.</i> <i>Sample Key Questions for Option 3 (after the presentations): Based on your K and W entries, what do you know so far about</i>
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OPTION 3: KWL CHART

Students write what they *Know* and *Want to know* about ecosystems and climate change.

Procedure:

1. Distribute KWL charts (K = What I Know, W = What I Want to Know, L = What I Learned).
2. Individually, learners fill in the K and W columns. The teacher collects some responses and lists them on the board to highlight misconceptions and interests. At the end of the lesson, learners will complete the L column for reflection.

Assessment/Rubric (5 pts each = 20 total):

Completeness of Entries

The teacher uses this activity to visually capture learners' prior knowledge and associations about ecosystems and climate change. By creating a collective mind map, students organize ideas, highlight connections, and surface both accurate information and misconceptions. This helps the teacher identify gaps in understanding and tailor the upcoming lesson more effectively.

Sample Key Questions for Option 5 (after the presentations): Based on the mind map, what do you know so far about the relationships between Philippine ecosystems and climate change?



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	Accuracy of Prior Knowledge (K) Relevance of Questions (W) Neatness/Organization OPTION 4: QUICK DEBATE <i>Prompt: “Which ecosystem in the Philippines is most vulnerable to climate change?”</i> Procedure: 1. The teacher divides class into two or more sides (e.g., marine vs. forest ecosystems).	
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2. Give learners 3 minutes to prepare simple arguments. Each side presents short points (30–60 seconds per student).
3. The teacher facilitates quick rebuttals and summarizes the discussion.

Assessment/Rubric (5 pts each = 20 total):

Strength of Argument

Relevance to Ecosystem/Climate Change

Team Participation

Respectful Delivery

OPTION 5: MIND MAP RECALL



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Groups draw a quick map of Philippine ecosystems and brainstorm possible climate threats.

Procedure:

1. Group learners (4–5 per team).
2. In the center of the paper, write “Philippine Ecosystems”. Students draw branches to show ecosystem types, then add sub-branches for possible climate threats (e.g., typhoon, drought, sea level rise).
3. Groups present their mind maps to the class in 1–2 minutes each.

Assessment/Rubric (5 pts each = 20 total):

Accuracy of Content

Organization of Mind Map

Creativity/Visual Appeal



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	Group Participation 2. Establishing the Purpose of the Lesson The teacher asks the learners: • “Why do you think we need to know and be aware of the present situation of the Philippine ecosystem?” • “How do you think climate change affects the ecosystems around us?” The teacher can also allow the learners to think of their dream job, career, or community role (e.g., farmer, fisherfolk, marine biologist, environmentalist, engineer, entrepreneur) and ask: • “How might changes in our ecosystems due to climate change affect that job or community role?” Then, the teacher presents the unit objectives for the learners to read.	
B. Instituting New Knowledge	1. Presenting Examples The teacher explains that learners already encountered the basic idea of ecosystems in earlier grade levels (such as food chains, habitats, and biodiversity). In this unit, they will	<i>If the learners are unable to provide the appropriate definitions of the difficult ecological terms (e.g., biodiversity, coral bleaching, habitat degradation), the teacher should step in to</i>



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revisit those ideas but will also encounter new and more complex concepts, such as ecosystem services, biodiversity loss, and climate change impacts that were not fully tackled before. The teacher shows examples of Philippine ecosystems (forest, coral reef, mangrove, agricultural land, freshwater river) together with their brief definitions and key features. The learners read them. The teacher notes that learners will later apply these concepts in the next activities to explain how climate change affects each ecosystem. For each ecosystem type, learners are given time to study the description and ask about words or terms they find difficult to understand (e.g., “biodiversity,” “habitat degradation,” “coral bleaching”). The teacher then invites other students to help explain the difficult terms in their own words. The teacher asks the learners if they are already familiar with some of these ecosystems. The teacher reemphasizes that while ecosystems were introduced in earlier grades, this unit will explore them in a deeper and broader way—specifically in relation to the climate crisis in the Philippines. ACTIVITY 1: INTERACTIVE LECTURE WITH VISUALS The teacher begins by presenting the major ecosystems in the Philippines—forests, coral reefs, mangroves, agricultural	<i>give clear and accurate explanations. Suggested definitions may include:</i> Biodiversity <i>refers to the variety of life in a particular habitat or ecosystem.</i> Habitat degradation <i>refers to the process by which a natural habitat becomes incapable of supporting its species.</i> Coral bleaching <i>refers to the whitening of corals due to stress factors such as warmer ocean temperatures.</i> <i>The teacher ensures that learners have a basic understanding of these concepts before moving forward. To check comprehension, learners may be asked to give examples of these terms in the context of Philippine ecosystems. The teacher may modify the examples or add more depending on the learners’ level of familiarity.</i>
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lands, and freshwater systems—using slides or infographics. For each ecosystem, the teacher highlights its role (for example, mangroves protecting coastlines) and its vulnerability to climate change (such as being destroyed by rising sea levels). Learners are given a guided worksheet with fill-in-the-blank notes, which they complete as the lecture progresses. After each ecosystem is introduced, the teacher pauses to ask simple comprehension questions and invites students to share examples from their own community. Learners are expected to participate actively, fill in their notes accurately, and clarify unfamiliar terms through class discussion.

Processing Questions:

1. Why is this ecosystem important to both the environment and human life?
2. In what ways does climate change specifically threaten this ecosystem?
3. How does the destruction of this ecosystem affect nearby communities?

ACTIVITY 2. VIDEO CLIP ANALYSIS

The teacher uses this activity to build foundational knowledge of Philippine ecosystems while engaging learners through visuals and guided notetaking. The pauses for questions allow learners to connect concepts to real-life contexts.

Possible answer to Processing Questions:

1. Learners may answer that loss of mangroves increases flooding risks or loss of reefs reduces fish catch.
2. Learners might mention providing food, shelter for wildlife, or livelihood for people.
3. Possible responses include rising sea levels destroying mangroves, coral bleaching reducing reef biodiversity, or droughts affecting agriculture.

These exchanges ensure misconceptions are addressed before moving deeper into climate change impacts.

The teacher can access this link:
<https://youtu.be/y19WEesgy2k>



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The teacher plays a short documentary or news report showing how climate change has affected a specific Philippine ecosystem, such as coral bleaching in Palawan, mangrove loss after Typhoon Yolanda, or drought in Mindanao's farmlands. Learners work in pairs to answer guide questions about the video: What ecosystem is featured? What visible signs of climate change are present? How does this affect people living nearby? Afterward, pairs share their answers in a whole-class discussion, and the teacher synthesizes the points by linking them to the day's objectives. Students are assessed on their ability to observe details, explain cause-effect relationships, and connect ecosystem impacts to human life.

Processing Questions:

1. What changes did you observe in the ecosystem shown in the video?
2. How are these changes connected to climate change?
3. What are the consequences of these changes for people living in that area?

ACTIVITY 3. CAUSE-AND-EFFECT DIAGRAM

Learners are divided into small groups, and each is assigned a climate change factor, such as sea level rise, stronger typhoons, or increasing temperature. Using Manila paper and markers, each group creates a flowchart showing the chain of

The teacher uses this activity to strengthen observation and analytical skills by having learners analyze real-world examples of climate change impacts on Philippine ecosystems.

Possible answer to Processing Questions:

1. *learners may answer coral reefs turning white (coral bleaching), mangroves uprooted by storm surges, or dried farmlands from drought.*
2. *Learners might mention rising sea temperatures, stronger typhoons, or reduced rainfall.*
3. *Reduced fish catch and loss of livelihood, greater flooding risks, food insecurity, or displacement of communities.*

These responses allow the teacher to highlight how ecosystems and human well-being are deeply interconnected under climate change.



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effects: the climate factor, its impact on an ecosystem, and its consequence for human communities. For example, rising temperature → coral bleaching → fish population decline → reduced income for fisherfolk. After completing the diagram, each group presents their work in front of the class in three minutes, explaining the links they drew. The teacher and classmates may ask clarifying questions. Learners are assessed based on accuracy, organization, teamwork, and clarity of explanation. Processing Questions: 1. What connections did you identify between climate factors and ecosystem impacts? 2. How do these ecosystems changes affect human activities like farming or fishing? 3. Which impact do you think is the most serious, and why? ACTIVITY 4. JIGSAW READING (PEER TEACHING) The teacher distributes short readings, with each one focusing on a different ecosystem—forest, marine, freshwater, agricultural, or urban. Students first meet in “expert groups,” where everyone has the same reading, and together they	<i>The teacher uses this activity to help learners visualize the chain of impacts from climate change to ecosystems and human life, strengthening their cause-and-effect reasoning.</i> <i>Possible answers to Processing Questions:</i> 1. <i>Learners may identify connections like sea level rise → mangrove loss → weaker coastal protection.</i> 2. <i>They might explain that ecosystem changes disrupt farming, fishing, or tourism.</i> 3. <i>Learners may consider the most serious impacts to be food insecurity, displacement of families, or loss of biodiversity.</i> <i>This enables the teacher to emphasize the interconnection between climate factors, ecosystem health, and human survival.</i>
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summarize the key points into three main ideas. Afterward, learners form “mixed groups,” each containing at least one expert on every ecosystem. In these new groups, students take turns teaching their peers what they learned from their reading while listening and taking notes on other ecosystems. The teacher moves around to check discussions, clarify misconceptions, and encourage equal participation. At the end, the teacher asks for volunteers to summarize the ecosystems covered and highlight how climate change impacts them.

Processing Questions:

1. What was the most important point you learned about your assigned ecosystem?
2. How did climate change affect this ecosystem differently compared to others?
3. After hearing from your groupmates, which ecosystem seems most vulnerable, and why?

ACTIVITY 5. CASE STUDY ANALYSIS

The teacher provides each small group with a printed case study describing a real Philippine scenario, such as Typhoon Yolanda’s impact on coastal communities, coral reef bleaching in Tubbataha, or prolonged drought in Mindanao affecting rice production. Each group reads their assigned case carefully and answers three guide questions: Which

The teacher uses this activity to promote collaboration and peer teaching, ensuring learners gain a broader understanding of how climate change impacts different ecosystems.

Possible answers to Processing Questions:

1. *Learners may note that forests regulate water and store carbon, while coral reefs support fisheries and tourism.*
2. *They might explain that typhoons devastate forests, while coral bleaching and rising seas primarily harm marine systems.*
3. *Learners may conclude that marine or coastal ecosystems are most vulnerable due to their exposure to multiple climate threats like rising seas and stronger storms.*

This allows the teacher to stress that while all ecosystems are interconnected, each faces unique challenges under climate change.

Allow the learners to make mistakes. Explain that mistakes are part of the process of learning and discovery.



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ecosystem was affected? How did climate change contribute to the situation? What were the short- and long-term consequences for both the environment and the people? After discussing among themselves, the groups present their analysis to the class in a short oral report. Other groups may ask questions or compare findings across different cases. Learners are assessed on their understanding of the case, accuracy of explanation, depth of analysis, and presentation skills. Processing Questions: 1. What evidence from the case shows the impact of climate change? 2. How did the affected ecosystem's condition influence human lives in the story? 3. What lessons can we learn from this case to prevent or reduce similar impacts in the future? Checkpoint: The teacher asks the learners what new terms and concepts they have learned so far based on the activities.	<i>The teacher uses this activity to strengthen critical thinking and problem-solving by connecting real community issues to climate change.</i> <i>Possible answers to Processing Questions:</i> 1. Drought → crop failure → income loss for farmers. 2. Flooding → waterborne diseases → displacement of families. 3. Suggested strategies: water conservation, mangrove planting, early warning systems. <i>The teacher uses this checkpoint to monitor learners' grasp of new vocabulary and concepts (e.g., ecosystem services, coral bleaching, biodiversity loss).</i> <i>Possible Responses:</i> <i>"I learned that coral bleaching happens when the water gets too hot."</i>
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	2. Discussing New Concept 1. The teacher posts a chart on the board with two columns: Types of Ecosystems in the Philippines and Impacts of Climate Change. 2. The teacher pulls out meta cards with ecosystem names (Forest, Coral Reef, Mangrove, Freshwater, River/Lake, Agricultural Land, Grassland, Urban Ecosystem) and impact terms (Deforestation, Coral Bleaching, Sea Level Rise, Soil Erosion, Flooding, Drought, Habitat Loss, Urban Heat Islands, Biodiversity Decline). 3. Learners are called one at a time to paste the meta cards in the correct column and try to match each ecosystem with one or more possible climate change impacts. Example: Coral Reef → Coral Bleaching Agricultural Land → Drought	<i>“Ecosystem services are the benefits people get from nature, like food, water, or flood protection.”</i> <i>“Biodiversity loss means fewer species are surviving because their habitats are destroyed.”</i> <i>“Mangroves are important because they protect communities from storm surges.”</i> <i>The teacher uses this matching activity to actively engage learners in linking Philippine ecosystems with the specific impacts of climate change. This strategy strengthens recall of ecosystem types while deepening understanding of cause-and-effect relationships. By encouraging learners to explain why certain impacts occur (e.g., coral bleaching due to rising sea temperatures, soil erosion from stronger typhoons), the teacher fosters critical thinking and application of scientific reasoning.</i> <i>Possible Responses:</i> <i>“Coral reefs bleach because higher temperatures stress the corals and make them lose their color.”</i>
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	Forest → Deforestation and Habitat Loss 4. After matching, the teacher asks a learner to explain why a particular ecosystem is vulnerable to the impact shown. For instance: “Why do coral reefs bleach when water temperature rises?” or “Why do stronger typhoons cause soil erosion in farms?” 5. If learners’ explanations are incomplete, the teacher provides additional clarification, reinforcing concepts with simple analogies or real events from the Philippines. 6. The teacher then asks the class to identify difficult terms from the cards (e.g., “coral bleaching,” “urban heat island,” “habitat loss”) and invites other learners to give their own definitions or examples before the teacher provides the correct explanation. 7. This process is repeated until all ecosystems and impacts are correctly matched, explained, and clarified. Activity 6: ECOSYSTEM IN A JAR (CLIMATE CHANGE IMPACTS) The teacher begins with an introductory statement: “Our ecosystems—forests, reefs, mangroves, rivers, and farmlands—are like living jars of life. Everything inside them is connected. But what happens when climate change disrupts these systems?” The teacher briefly explains that	<i>“Forests suffer from deforestation when people cut trees, and this leads to habitat loss.”</i> <i>“Agricultural lands are vulnerable to drought because crops need water to survive.”</i> <i>“Urban areas experience heat islands because of too many buildings and less trees.”</i>
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learners will simulate how climate change affects ecosystems using a simple group activity. The teacher asks the learners to form small groups of 4–5. The teacher provides the materials: a transparent container (jar or plastic bottle), soil, water, small plants or leaves, and slips of paper labeled with “climate change events” such as heavy rainfall, drought, typhoon, sea level rise, extreme heat. The teacher instructs learners to first set up a mini “ecosystem in a jar” by placing soil, water, and plants. Once prepared, each group will randomly pick one climate change event card. The group then simulates how this event would affect their mini ecosystem. For example, if the card says drought, they reduce the water and observe how the soil and plants respond. If the card says heavy rainfall, they pour excess water and see flooding effects. After the simulation, the teacher facilitates class discussion using the following processing questions: How did your mini ecosystem change after the climate event? Which part of the system was most vulnerable, and why? How do these changes mirror what happens in real Philippine ecosystems? What can people do to reduce the impacts of these changes in their communities?	<i>The teacher emphasizes that ecosystems are fragile and interconnected: a change in one factor can disrupt the whole system.</i> <i>Possible Answers:</i> • <i>Soil dried and plants wilted (drought).</i> • <i>Flooding damaged the roots (heavy rainfall).</i> • <i>Plants overheated and wilted faster (extreme heat).</i> • <i>These mirror real Philippine ecosystems like farmlands during drought or coral reefs during warming.</i> • <i>Communities can help by planting trees, conserving water, and preparing for disasters.</i>
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The teacher emphasizes that ecosystems are fragile and interconnected: a change in one factor (like rainfall or temperature) can disrupt the whole system. The activity helps learners visualize how climate change impacts forests, reefs, farmlands, and other ecosystems in the Philippines.

Now, the learners are expected to understand how climate change directly affects ecosystems. The teacher states that they will engage in more activities to deepen their mastery of the topic.

3. Developing Mastery

ACTIVITY 7 (OPTION 1): CLIMATE CHANGE IMPACT PUZZLE

The teacher prepares puzzle pieces where each piece contains either an ecosystem (forest, coral reef, mangrove, freshwater, agricultural land) or an impact of climate change (deforestation, coral bleaching, flooding, drought, soil erosion). Learners are grouped into teams of 4. Each team receives a set of shuffled puzzle pieces. Their task is to match each ecosystem with its correct climate change impact(s)



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	within a limited time. Once the puzzle is completed, the teacher calls on groups to justify their matches (e.g., “We matched coral reefs with coral bleaching because warmer waters damage corals.”). The teacher clarifies misconceptions and highlights correct connections.	<i>The teacher uses this puzzle activity to reinforce learners’ ability to connect ecosystems with specific climate change impacts in an interactive and engaging way. It strengthens recall, critical thinking, and teamwork while allowing misconceptions to surface for clarification.</i> <i>Possible Answers:</i> <i>Coral reefs → coral bleaching due to warmer waters.</i> <i>Forests → deforestation and habitat loss from logging and typhoons.</i> <i>Mangroves → flooding and storm surges destroying coastal protection.</i> <i>Agricultural land → drought reducing crop yield.</i> <i>Freshwater → soil erosion and flooding affecting rivers and lakes.</i>
	ACTIVITY 7 (OPTION 2): ROLE-PLAY “VOICE OF THE ECOSYSTEM” 1. The teacher assigns each small group of learners one Philippine ecosystem (forest, reef, mangrove, agricultural land, or river). 2. Each group prepares a short 2–3 minute role-play where the ecosystem “speaks” about its condition under climate change. For example, the coral reef may say: “Because of rising temperatures, I am bleaching, and the fish that depend on me are dying.”	<i>Learners act as representatives of different ecosystems (forest, coral reef, mangrove, freshwater, agricultural). Each group presents how climate change affects them, speaking as if they were the ecosystem itself.</i>



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3. After each role-play, the teacher asks the class to identify the key climate change impact and how it affects both the ecosystem and people.
4. The teacher summarizes the activity by emphasizing that ecosystems are “living systems” that show visible effects of climate change.

ACTIVITY 7 (OPTION 3): GALLERY WALK – BEFORE AND AFTER

1. The teacher posts images around the classroom showing Philippine ecosystems in two conditions: healthy and impacted by climate change (e.g., a lush mangrove vs. a damaged one after a typhoon, a vibrant coral reef vs. a bleached one).
2. Learners walk around in pairs, observing the pictures. They are given a worksheet with guiding questions: What ecosystem do you see? What signs of climate change are present? How might this change affect local communities
3. After the walk, pairs share their observations, and the teacher collects key insights on the board, grouping them by ecosystem.
4. The class discusses patterns across the images, such as how all ecosystems show vulnerability to extreme weather, rising seas, or warming.

The teacher applies role-play to encourage empathy and creativity. By letting learners “speak” as ecosystems, the activity fosters deeper connections to biodiversity and strengthens their ability to communicate environmental issues.

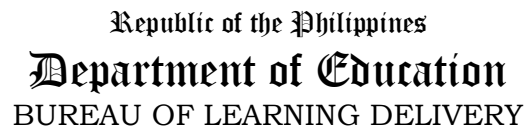
The teacher uses the gallery walk to encourage active observation and peer interaction. Walking around in pairs helps learners engage with the visual contrasts of ecosystems before and after climate change impacts. Sharing insights fosters collaboration, while grouping responses on the board allows the class to see common patterns. This activity strengthens visual analysis skills and highlights the widespread vulnerability of ecosystems.



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		<i>Possible Answers (sample insights):</i> • Mangrove: damaged by typhoon → reduced coastal protection, flooding risk for nearby homes. • Coral reef: bleaching → loss of fish habitat, decline in fisherfolk income. • Forest: deforestation after storm → soil erosion, landslides threatening communities. • Agricultural land: drought → lower crop production, food insecurity. • Freshwater river: polluted or dried → reduced drinking water and irrigation supply.
C. Demonstrating Knowledge and Skills	The teacher explains that the ultimate purpose of the lesson is for the learners to understand not only the scientific concepts of ecosystems and climate change but also their real-world significance. These ideas are directly connected to the learners' daily lives—whether in farming, fishing, tourism, health, or community safety—and are equally important in preparing them for their senior high school exit pathways, future careers, and civic responsibilities. Thus, the following additional activities will expose them to practical applications of how climate change affects Philippine ecosystems and human communities. 1. Finding Practical Applications ACTIVITY 8 OPTION 1: LOCAL NEWS ANALYSIS	<i>This activity emphasizes the relevance of classroom learning to real-life issues faced by Filipino communities. By analyzing local news articles, learners connect scientific concepts of climate change and ecosystems to everyday experiences such as farming, fishing, and community safety. To ensure inclusivity, the teacher may allow learners without internet access to bring</i>

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	Learners share their reflections in pairs, then volunteers share with the whole class. The teacher emphasizes that understanding ecosystems is not only academic—it's connected to real future opportunities. ACTIVITY 9 OPTION 1: “CLIMATE CHANGE TALK SHOW” The class is divided into groups of 6. Each group creates a short role-play formatted as a talk show. Roles may include a host, climate scientist, farmer, fisherfolk, government official, and student. Each person must explain their perspective on how climate change affects ecosystems and communities. The group presents their skit, and afterward, the teacher highlights the key insights and overlaps.	<i>will likely realize that solutions are complex, requiring both environmental and social considerations.</i>
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- c. **Venn Diagram** – Compare how two ecosystems (e.g., forest and mangrove) are similarly and differently affected by climate change.
 - d. **Table or Chart** – List ecosystems, the specific climate change impacts, and their effects on biodiversity and people.
3. Arrows should indicate the **direction of influence** (e.g., cause → effect). Use short phrases or symbols to clarify the connections.
 4. Learners work in **pairs or small groups** to complete their maps. Each group will then explain the reasoning behind their chosen links and structure.
 5. The teacher will facilitate a short discussion highlighting the **interconnectedness** among climate change, ecosystem health, and human survival—emphasizing the chain:

Climate Change → Ecosystems → Biodiversity →
People’s Livelihood and Well-being

ACTIVITY 10: INTRODUCTION TO ECO-FRIENDLY SOLUTIONS

1. The teacher shows a short video on “*How Climate Change Affects Philippine Ecosystems and What We Can Do*” (links can be replaced with any documentary



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	or local DENR/NGO resource video). The video highlights how mangroves, coral reefs, forests, and agricultural lands are being affected by rising temperatures, flooding, and biodiversity loss, while also presenting community-based adaptation practices such as mangrove planting, organic farming, and coastal cleanups. - The learners are tasked to define in their own words “climate change adaptation” and “climate change mitigation.” Their answers are discussed in class to build a shared understanding. - The teacher then shows the learners a Philippine infographic or data set (e.g., from PAGASA, DENR, or WWF-Philippines) about the impacts of climate change on ecosystems and the cost of inaction (flood damages, coral reef loss, agricultural decline). The class discusses the data with these guide questions: - Why do you think climate change has such a large impact on Philippine ecosystems, and what does this suggest about the urgency of local action? - How can community-based adaptation projects (like mangrove reforestation or organic farming) help both ecosystems and human communities? - What industries or sectors in the Philippines (e.g., tourism, fisheries, agriculture) are most vulnerable to ecosystem loss due to climate change, and why?	<i>They can access this link: “Philippines: Changing Climate Changing Lives” (short document: https://youtu.be/VvmPRrJC5cc</i> And from, Born to be Wild: Effects of deforestation in the Philippines link: https://youtu.be/em1eRDsXD4U <i>The teacher uses this activity to connect scientific concepts with real-life community practices. By analyzing the video and infographic, learners see how protecting ecosystems is not only about nature but also about jobs, food, and safety. This helps</i>
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	• How can young people like you take part in adaptation and mitigation practices in your schools, homes, and barangays? • How does protecting ecosystems also protect jobs, food supply, and future careers in the Philippines? ACTIVITY 11 OPTION 1: VIDEO ANALYSIS ON A LOCAL ADAPTATION PROJECT The teacher shows to the class a 3–5-minute video (e.g., DENR, WWF, or a YouTube documentary) about a Philippine-based climate adaptation project such as <i>mangrove reforestation in Leyte</i>, <i>coral reef rehabilitation in Palawan</i>, or <i>agroforestry in Bukidnon</i>. The video may be played more than once. Before the video is played, the learners are advised to note how the project addresses ecosystem threats and benefits local communities. The following guide questions are written on the board: • How does this project help reduce the impact of climate change on the ecosystem? • How does this project also benefit the people (jobs, food, protection from disasters)? • What challenges might the project face in the future?	<i>them realize their role in adaptation and mitigation as students and future professionals.</i> <i>Possible Learner Responses</i> • “The Philippines is very exposed to typhoons and rising seas, so our ecosystems are easily damaged. That’s why we need to act fast.” • “Mangrove planting helps stop flooding and also gives people in coastal areas more fish to catch.” • “Tourism and fisheries are most at risk because they depend on reefs and seas. If these are gone, jobs are lost.” • “As students, we can join cleanup drives, plant trees, and start waste reduction at school or home. Protecting nature also protects our future.”
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	- If you were part of this community, how would you improve or support this project? After watching, learners work in pairs to answer the questions. The teacher then facilitates a class discussion, encouraging different groups to share and compare their insights. ACTIVITY 11 OPTION 2: PERSONAL READINGS ON LOCAL ECO INITIATIVES The teacher assigns learners to research (online or from local news) a real adaptation or mitigation project in the Philippines. Examples: <i>Bantay Dagat</i> coastal protection groups - Community mangrove planting - School-based waste segregation programs - Organic farming initiatives - Coral reef restoration projects Each learner brings a short description or printed photo of the project. They answer the following questions: - What is the project about? - How does it help the ecosystem? - How does it help the people?	<i>This is the link of the video: “Philippines’ most successful mangrove reforestation project!”</i> https://www.youtube.com/watch?v=53zh0ZnY0g <i>Possible Learners Responses:</i> <i>“Mangrove planting reduces flooding and protects coasts from strong waves.”</i> <i>“The project gives people new jobs and more fish, so it helps the community, too.”</i> <i>“Future challenges might be cutting trees again or lack of government support.”</i> <i>“If I were part of the community, I would help maintain the mangroves and teach others why they are important.”</i>
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	4. What makes this project different from other local initiatives? 5. What improvements can you suggest? Learners then share their findings with a partner, and the teacher calls on volunteers to present to the class. The discussion is processed to emphasize that adaptation and mitigation efforts are happening in real communities, not just theory. 2. Making Generalization ACTIVITY 12 OPTION 1: COMEDIC SKITS The class is divided into groups of five or six. Each group creates a short skit showing a community scenario where ecosystems are being affected by climate change (e.g., flooding, reef bleaching, forest fire, agricultural decline). • In the skit, characters must mention scientific terms learned (ecosystem, biodiversity, climate change impact, adaptation, mitigation). • The story must also highlight a solution (e.g., planting mangroves, switching to organic farming, practicing waste segregation, protecting coral reefs). • Presentations are performed in front of the class.	<i>Learners can access the following links to read:</i> • Bantay Dagat coastal protection groups https://www.theguardian.com/environment/2024/mar/05/if-we-cant-fish-any-more-were-going-to-die-the-volunteer-sea-patrols-protecting-precious-philippine-waters • Community mangrove planting https://www.wetlands.org/case-study/scaling-science-based-mangrove-restoration-in-the-philippines/ • School-based waste segregation programs https://www.emb.gov.ph/solid-waste-management-programs/ • Organic farming initiatives https://ati2.da.gov.ph/ati-main/programs/organic-agriculture • Coral reef restoration projects https://globalfundcoralreefs.org/reef-
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	After the role plays, the teacher asks: • What common challenges did all the groups show? • What solutions were highlighted? • How does this activity connect to our own local community? ACTIVITY 12 OPTION 2: CONCEPT MAP The teacher provides learners with cards or a list of key terms: <i>ecosystem, biodiversity, coral reefs, forests, mangroves, agriculture, climate change, adaptation, mitigation, flooding, reef bleaching, food security, community projects</i>. The learners draw a concept map linking these terms. They must explain why they connected each idea (e.g., “Mangroves → flood protection,” “Reef bleaching → loss of biodiversity → loss of fish catch”). Learners submit their work, and the teacher summarizes by showing a master concept map that connects ecosystems, impacts of climate change, and adaptation/mitigation measures. 3. Evaluating Learning	<u>plus/news/women-rangers-on-the-front-lines-of-mangrove-and-coral-reef-protection-in-the-philippines/</u> <i>Learners may identify projects such as mangrove planting in Leyte, Bantay Dagat patrols protecting marine resources, or school-led zero waste programs.</i> <i>They might note ecosystem benefits (e.g., restored habitats, cleaner coasts, improved biodiversity) and human benefits (e.g., better fish catch, reduced flooding, healthier communities).</i> <i>Learners may suggest improvements like stronger funding, more youth involvement, or replication of projects in other areas.</i> <i>Possible Outcomes (answers):</i> <i>Common challenges: flooding in lowland areas, coral reef bleaching, soil erosion, forest loss, declining crop yield.</i> <i>Solutions highlighted: mangrove planting, organic farming, coral reef protection, tree planting, waste segregation, disaster preparedness.</i> <i>Connection to local community: these issues reflect real problems such as typhoon damage, loss of fish catch, urban</i>
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ACTIVITY 13: SHORT RESPONSE

Learners individually answer the following in their notebooks (words, phrases, or short sentences are acceptable):

1. What is an ecosystem, and why are Philippine ecosystems important?
2. Name **two impacts of climate change** on Philippine ecosystems.
3. Give one example of an adaptation project and a mitigation project.
4. How do healthy ecosystems protect people from the effects of climate change?
5. As a student, what simple action can you do to help protect ecosystems from climate change?

The teacher collects and checks the answers. Learners with incomplete or confused responses may be asked to pair with classmates who got the correct answers for peer clarification before final checking.

The following activity is optional and may be conducted if there is extra time beyond the teacher's allocated budget for this unit or lesson. Since it directly applies the concepts of climate change impacts on Philippine ecosystems, it may also serve as a meaningful performance task. This allows learners to extend their understanding by connecting classroom

flooding, or heat waves—showing that protecting ecosystems also protects people's health, livelihood, and safety.

Possible Outcomes:

Learners show connections such as: sea level rise → mangrove flooding → reduced fish nursery → loss of livelihood.

Common themes: all ecosystems are vulnerable; human survival depends on healthy ecosystems.



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	knowledge to real-world environmental issues and community actions. 4. Additional Activities ACTIVITY 14 OPTIONAL: DREAM COMMUNITY ECOSYSTEM - The teacher distributes a sheet of paper to each learner. - The teacher instructs the learners to imagine their dream sustainable community in the Philippines—one that is resilient to climate change and protects local ecosystems (e.g., mangroves for flood protection, eco-parks, organic farms, coral reef tourism, renewable energy). - Learners must: Draw or sketch their dream community ecosystem. Write at least three sentences explaining how the community protects biodiversity and adapts to climate change impacts (e.g., “Our community plants mangroves to reduce storm surges,” or “We practice organic farming to keep the soil healthy.”). - Indicate at least one career or business opportunity that could emerge in their dream community (e.g., eco-	<i>Possible Outcomes:</i> <i>Learners may write: “Coral reefs are important because they provide fish habitats and protect coasts. Climate change causes bleaching, which reduces fish catch and income for fisherfolk. Planting mangroves and reducing pollution can help communities adapt.”</i> <i>Answers show if students can connect ecosystems, impacts, and solutions in their own words.</i>
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	tourism guide, organic farm entrepreneur, marine biologist, renewable energy engineer). The teacher and class agree on the rubric, considering these suggested criteria: • Clarity of explanation (40%) • Creativity and completeness of sketch (40%) • Connection to climate change adaptation/mitigation (20%) After completing the activity, learners share their outputs with a partner. Selected learners are then called to present their dream community ecosystem to the class.	<i>Possible Outcomes:</i> <i>Learners may present communities with solar-powered schools, mangrove-protected coasts, organic farms, bike lanes, and waste segregation systems.</i>
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V. ASSESSMENT <i>(Assessing Learnings)</i>	Two options are given in this part: a performance-based assessment and a paper-and-pen assessment. The teacher may opt to implement both at his/her discretion and based on availability of time and the needs of the learners. ASSESSMENT OPTION 1: CLIMATE CHANGE IMPACT EXHIBIT — “Stories from the Earth” As a culminating performance task, learners will work collaboratively to create a mini-exhibit that showcases real-world examples of how climate change affects various Philippine ecosystems and human communities. This activity allows them to analyze local climate issues and demonstrate understanding of the interconnected relationship between climate change, ecosystems, and human life. Learners are grouped into teams of 4–6 members. Each group selects one Philippine ecosystem (e.g., mangrove forest, coral reef, upland forest, grassland, or agricultural area) and research specific climate change impacts observed in that setting—such as coral bleaching, flooding, drought, biodiversity loss, or soil erosion. They also gather or create materials (photos, short narratives, infographics, or charts) showing how these impacts affect both the environment and people’s livelihoods.	



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Each group will then organize their findings into a visual exhibit panel (poster, infographic board, or digital presentation) titled “Stories from the Earth: Climate Change and Our Ecosystems.” The exhibit must include:

- The chosen ecosystem and its role in supporting biodiversity and communities.
- Real examples of climate change impacts affecting it.
- How these changes influence people’s lives, livelihoods, and health.
- A brief reflection on what can be done to strengthen resilience or awareness in their community.

The class may hold a “gallery walk” where groups display and explain their panels. Visitors (classmates or other groups) can ask questions or give feedback on clarity, accuracy, and insight. The teacher concludes by synthesizing how all exhibits show the interconnectedness between climate, ecosystems, and human life.

The teacher may use and modify the following rubric after presenting it to the class.

Criteria	Points
Content Accuracy and Relevance	10 points
Analysis and Understanding	10 points
Organization and Visual Presentation	10 points
Collaboration and Participation	10 points
Reflection and Insight	10 points
Total:	50 points



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ASSESSMENT OPTION 2: PAPER-AND-PEN ASSESSMENT (SHORT RESPONSE TEST)

Learners answer the following questions individually in their notebooks or test papers:

1. Define an **ecosystem** in your own words and give one example in the Philippines.
2. Identify **two impacts of climate change** on Philippine ecosystems.
3. Explain briefly how **coral reefs or mangroves** are affected by climate change.
4. Suggest one **adaptation strategy** and one **mitigation strategy** for protecting ecosystems.
5. Why is protecting Philippine ecosystems also important for local communities and future careers?

Scoring: Each item is worth 2 points (10 points total). The teacher may use the following guide:



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- 2 points = Complete and accurate answer
- 1 point = Partial or unclear answer
- 0 points = Incorrect or no answer

ASSESSMENT OPTION 3: MULTIPLE CHOICE TEST

Directions: Read each item carefully. Choose the letter of the correct answer.

1. Which of the following best describes an ecosystem?
 - a. A collection of plants only.
 - b. A single species in each location.
 - c. A group of animals living in the same area.
 - d. A community of organisms interacting with each other and their physical environment.
2. Which Philippine ecosystem serves as the “nursery grounds” for many fish species but is threatened by coral bleaching due to rising sea temperatures?
 - a. Forests
 - b. Mangroves
 - c. Coral reefs
 - d. Grasslands
3. Which of the following is not a direct impact of climate change on ecosystems?
 - a. Coral bleaching
 - b. Sea level rise
 - c. Volcanic eruption
 - d. Increase in storm intensity



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| | <ol style="list-style-type: none">4. What role do mangroves play in climate change adaptation?<ol style="list-style-type: none">a. They speed up soil erosion.b. They provide shade for tourists.c. They prevent landslides in mountain areas.d. They trap carbon dioxide from the atmosphere.5. Which Philippine ecosystem is most threatened by deforestation and illegal logging, worsening the effects of climate change?<ol style="list-style-type: none">a. Coral reefsb. Forestsc. Mangrovesd. Grasslands6. Which of the following best describes coral bleaching?<ol style="list-style-type: none">a. Corals producing more fish larvaeb. Corals increasing their biodiversity.c. Corals turning black due to pollution.d. Corals lose their color because of stress from heat or acidity.7. Which sector in the Philippines is most affected when rice fields are damaged by floods and droughts caused by climate change?<ol style="list-style-type: none">a. Educationb. Tourismc. Agricultured. Manufacturing8. Sea level rise poses the greatest threat to which of the following?<ol style="list-style-type: none">a. Inland rice farms.b. Mountain forests.c. River systems in upland areas.d. Coastal communities and mangrove ecosystems.9. Which of the following ecosystems acts as a natural “barrier” against storm surges and tsunamis?<ol style="list-style-type: none">a. Grasslands |
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| | <ul style="list-style-type: none">b. Urban parksc. Agricultural landsd. Coral reefs and mangroves <p>10. Which greenhouse gas is most associated with global warming and climate change?</p> <ul style="list-style-type: none">a. Heliumb. Nitrogenc. Oxygend. Carbon dioxide <p>11. A fisherman in Palawan notices fewer fish in the coral reefs due to bleaching. This is an example of:</p> <ul style="list-style-type: none">a. Air pollution causing less tourism.b. Deforestation impact on communities.c. Climate change has no effect on human activity.d. Economic impact of climate change on ecosystems. <p>12. Which Philippine ecosystem is most vulnerable to droughts caused by El Niño?</p> <ul style="list-style-type: none">a. Mangrovesb. Coral reefsc. Forest ecosystemsd. Agricultural ecosystems (farmlands) <p>13. Which of the following adaptation strategies can communities implement to protect ecosystems?</p> <ul style="list-style-type: none">a. Planting mangroves to reduce flooding.b. Cutting down trees for more housing projects.c. Building more concrete roads in coastal areas.d. Using more fossil fuels for faster economic growth. <p>14. Which of the following is an example of mitigation rather than adaptation?</p> <ul style="list-style-type: none">a. Installing early warning systems for floods.b. Planting mangroves to shield coasts from storms.c. Relocating communities away from flood-prone areas.d. Using solar panels to reduce greenhouse gas emissions. |
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	15. Why are Philippine ecosystems especially vulnerable to climate change? - The Philippines has no biodiversity. - The Philippines is located far from oceans. - Filipinos do not depend on ecosystems for survival. - The country is an archipelago with high exposure to typhoons and sea level rise.
VI. REFLECTION <i>(Feedback and Continuous Improvement)</i>	Learners' Reflection <i>Ask the learners to complete the following prompts in a piece of paper. They may write as many points as possible.</i> 1. In this unit, the most significant lesson I learned about Philippine ecosystems and climate change is _____. I will apply what I learned by _____ (e.g., supporting local environmental programs, conserving resources at home, raising awareness in my community). I am still confused about _____ (e.g., specific impacts of climate change on certain ecosystems, difference between adaptation and mitigation). My learning experience about ecosystems and climate change could have been better if _____. <i>Teacher's Reflection</i> - Were the objectives achieved? What evidence do I have that learners can now explain how climate change impacts Philippine ecosystems? - What went well in this lesson? (e.g., learner engagement in activities, strong connections made between scientific concepts and local realities). - What challenges did I encounter? (e.g., limited time for group work, learners' difficulty connecting global climate issues to local ecosystems). How did I handle them?



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	4. How can I improve my teaching practice to better support learners' understanding of how climate change affects ecosystems, and how they can take action in their communities?
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Annex A. **DESIGN THINKING PROCESS**

Project Title	
Project Proponents	



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Project Duration	
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What knowledge, skills, and attitudes (KSAs) are needed in this project?

Knowledge	Skills	Attitudes

EMPATHIZE Understand the context and the need. What glaring need or problem have you observed? Why is this a problem?	
DEFINE Define the scope of your problem. What problem do you commit to address?	
IDEATE Think of potential solutions. What are your ideas on how the problem will be solved?	Will these ideas work? How?
Idea 1 	
Idea 2 	
Idea 3 	



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PROTOTYPE/PRODUCE Create your solution. How would you translate your idea into a product, process, or output?	
TEST AND IMPLEMENT Implement your solution. How would you test if your idea is working? How would you roll out the solution?	
LEARN Evaluate your learning experience. What were your essential learning in the process of designing? Answer by highlighting the KSAs. How would this project impact you and the community if implemented? How can the project be improved?	

Annex B. **RATIONALIZATION OF THE MULTIPLE-CHOICE ASSESSMENT**



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Item No.	Correct Answer	Rationalization
1	D	An ecosystem is not just a group of plants or animals; it is a community of organisms interacting with each other and their environment .
2	C	Coral reefs serve as nursery grounds for fish. Rising sea temperatures cause coral bleaching , threatening marine biodiversity and fisheries.
3	C	Volcanic eruptions are natural disasters but not directly linked to climate change. Sea level rise, stronger storms, and coral bleaching are climate-driven.
4	D	Mangroves trap carbon dioxide and act as carbon sinks, helping mitigate climate change. They also provide coastal protection.
5	B	Forests are threatened by deforestation and illegal logging, which worsen biodiversity loss and reduce carbon absorption.
6	D	Coral bleaching occurs when corals lose their color due to stress from heat or acidity , leading to reef damage.
7	C	The agriculture sector is most affected when farmlands are flooded or experience droughts, threatening food security.
8	D	Coastal communities and mangroves are most vulnerable to rising seas, leading to flooding, erosion, and displacement.
9	D	Coral reefs and mangroves act as natural barriers that reduce wave energy and storm surge damage.
10	D	Carbon dioxide is the main greenhouse gas driving global warming. Oxygen, nitrogen, and helium do not cause this effect.
11	D	The loss of fish catch due to coral bleaching is an economic impact of climate change, directly affecting livelihoods.
12	D	Agricultural ecosystems (farmlands) are most vulnerable to droughts caused by El Niño, reducing crop yields.
13	A	Planting mangroves is a community adaptation strategy that reduces flooding and supports ecosystems.



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14	D	Using solar panels reduces greenhouse gas emissions, which is a mitigation strategy , while the others are adaptation.
15	D	The Philippines is an archipelago highly exposed to typhoons and sea level rise , making its ecosystems very vulnerable to climate change.